Magnet Poles — Attract & Repel

Grade 2 STEM Lesson • Thynk Unlimited

1. What Is a Magnet?

A magnet is a special object that can pull some things toward it. Magnets come in many shapes and sizes. They can attract objects made of iron and steel — like paper clips and nails. Magnets are truly magical objects in science!

■ Magnets pull things close — like invisible magic hands!

2. Types of Magnets

Magnets come in many different shapes and sizes. Each type has a North Pole and a South Pole at its ends.

Three Common Shapes:

- **Bar Magnet:** Long and straight — the most common magnet shape.
- **Horseshoe Magnet:** Curved like the letter U — very strong poles!
- **Disc Magnet:** Small and round — found in toys and gadgets.

3. What Are Magnet Poles?

Every magnet has two ends called poles. One end is the North Pole and the other end is the South Pole. The poles are where the magnet's pulling power is the strongest!

The North Pole and South Pole are at opposite ends of a magnet.

Every magnet always has BOTH a North Pole and a South Pole!

4. Meet the Poles: North & South

Every magnet has a North Pole (N) and a South Pole (S). The North Pole is usually red and the South Pole is usually blue. Each pole has its own special power!

■ Fun Fact: Poles are like magnets' special friends!

5. Opposites Attract!

When the North Pole of one magnet meets the South Pole of another magnet, they pull each other close! This is called **attraction**.

> Opposite poles pull each other close!

> **North Pole + South Pole = They attract!**

6. Likes Repel!

Same poles push away from each other! When two North Poles or two South Poles face each other, they push apart with a strong force.

> North + North = Push away from each other!

> **Same poles always repel — they never stick together!**

7. How Do Poles Work?

Magnets can pull or push depending on their poles.
• When **opposite poles** (North & South) face each other, they **ATTRACT** — they pull close together!
• When the **same poles** (North & North or South & South) face each other, they **REPEL** — they push away!

> Pull together or push apart — it all depends on the poles!

8. What Can Magnets Attract?

Magnets attract some metals like iron and steel. Here are some things a magnet can pull toward it:

- **Paper clips** — made of metal, they stick right to a magnet!
- **Iron nails** — iron is attracted to magnets easily.
- **Fridge magnets** — they stick because of the metal surface!

9. What Magnets Don't Attract

Magnets do not attract everything — like wood or plastic. Not all materials are magnetic!

- **Wood** does not stick to a magnet — try a wooden block!
- **Plastic toys** won't be pulled — plastic is not magnetic.
- **Paper** floats free — magnets cannot pick up paper!

10. Simple Magnet Experiment

Try these fun steps at home or in class! Explore how magnets attract and repel using simple objects around you.

1. Use a magnet to pick up paper clips.
2. Push two magnets together and pull apart.
3. See which poles attract or repel!

11. Magnets in Our Daily Life

Magnets are all around us! We use them every day in many helpful ways. Can you spot a magnet in your home?

Refrigerator magnets hold notes and drawings on the fridge!	**A compass** uses magnets to help us find directions.
Toys with magnets snap together for fun building play!	**Door latches** use magnets to keep doors closed gently.

Check Your Understanding

Name: ______________________________ Date: _______________

1. Every magnet has two ends called __________.

2. The two poles of a magnet are the __________ Pole and the __________ Pole.

3. When opposite poles meet, magnets __________ (attract / repel). Circle one.

4. When the same poles meet, magnets __________ (attract / repel). Circle one.

5. Name one object a magnet CAN attract: ______________________________

6. Name one object a magnet CANNOT attract: ______________________________

7. Draw a bar magnet below and label its North and South Pole.

■ **You Did Amazing! Keep asking, keep trying, keep discovering!** ■